Sandeep Kumar Jha

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.S-Economic Sciences	Indian Institute of Technology Kanpur	7.2/10
2022	Class XII (CBSE)	Progressive Central High School	88.4%
2020	Class X (CBSE)	Progressive Central High School	94.2%

Work Experience

Backend Developer | **Elewin.in** *Apr'26 – Jul'26*

- Engineered scalable **REST APIs** using **Node.js**, **Express**, and **MongoDB** for resume analysis and job-fit matching
- Implemented **JWT-based authentication** and **RBAC** through reusable **Express middleware** protecting API routes
- Implemented asynchronous **resume-processing workflows** using **BullMQ** and **Redis** with worker execution and retries
- Integrated **Google Gemini API** for automated **resume analysis** and **job-description comparison**, generating insights
- Optimized backend performance through **Redis caching**, **MongoDB indexing**, validation, and centralized error handling

AI Tools & Full Stack Developer Intern | **DigitX** *Apr'25 - May'25*

- Integrated AI image generation using **OpenAI DALL·E 3**, **Piapi**, and **Flux API** for dynamic content creation
- Implemented response interruption using **AbortSignal**, allowing users to stop AI generation during processing seamlessly
- Built AI-powered **Chart** and **Diagram** generation using **Chart.js** and **Mermaid** for interactive data visualizations
- Developed AI-powered **Video** generation using **Kling** and **PixVerse APIs** for text-to-video content creation workflows

Key Projects

Semantic Video Compression Pipeline | **Computer Vision** | **Self** | *Feb'26*

- Built video compression using **OpenCV**, **Dense Optical Flow**, **Haar Cascade**, and **pHash** for activity preservation
- Optimized processing with disk streaming, temporal downsampling, and spatial scaling, achieving **14× real-time** throughput
- Reduced storage requirements by **99.1%** (**158 MB → 1.4 MB**) while preserving essential semantic information
- Implemented adaptive motion-threshold calibration using the first **30 seconds**, improving robustness across noise and lighting

Codeforces Recent Action | **Programming Club, IIT Kanpur** | *Feb'24*

- Architected a web server using **Go** and **MongoDB** for structured storage and efficient querying of Codeforces blog updates
- Integrated concurrent data fetching using **Goroutines** and background workers to poll the Codeforces API without blocking
- Achieved **10× faster** data retrieval by optimizing concurrent request handling and background task execution for API polling

In-Memory Redis-like Key Value Store | **Self Project**| *Aug'26*

- Built an in-memory key-value store using **unordered_map** and **unordered_set**, supporting **O(1)** average operations
- Designed **Command Parser, Storage Engine, Expiry Manager,** and **Database** components using **OOP** principles
- Implemented key expiration using **C++ chrono** and file persistence for database recovery across application restarts

Credit Card Fraud Detection | **Self Project** | *Aug'26*

- Built a **leakage-safe** fraud pipeline on **284K+** transactions using **stratified cross-validation** and imbalance handling
- Compared Logistic Regression, Decision Tree, KNN, and **Random Forest** using **PR-AUC**, ROC-AUC, precision, recall, F1
- Tuned **thresholds** on validation data to optimize the precision-recall tradeoff, achieving **87% precision** and **74% recall**
- Analyzed confusion matrices and **feature importance** to quantify false-positive/false-negative tradeoffs and explain behavior

Technical Skills

Programming Languages	C++, C, Python, JavaScript, TypeScript, Go, SQL, Dart, LaTeX
Backend & Databases	Node.js, Express.js, MongoDB, Redis, BullMQ, Firebase, Cloud Firestore, REST APIs
AI/ML & Computer Vision	Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Statsmodels
AI & APIs	Google Gemini API, OpenAI API, DALL-E 3, Flux API, Kling API, PixVerse API
Frontend & Mobile	React, Next.js, Flutter, Chart.js, Mermaid
Developer Tools	Git, GitHub, Docker, VS Code, Firebase Storage, Firebase Auth, Razorpay, ObjectBox
Core Concepts	Data Structures & Algorithms, OOP, REST API Design, Authentication & Authorization, Caching, Asynchronous Processing, Message Queues, System Design

Relevant Courses

Fundamentals of Computing	Probability and Statistics	Data Structures & Algorithms

Positions of Responsibility

Web & App Manager | Inter IIT Sports Meet 2024 (*Sep'24 – Dec'24*)

- Developed and deployed the official **Inter IIT Sports Meet 2024** app, serving **5,000+ participants**
- Built secure **QR-based** meal verification, enabling real-time participant validation across multiple venues
- Developed APIs and database workflows for registration, QR verification, and event operations using **Node.js** and **Firebase**
- Led a **5-member engineering team** delivering networking, live score updates, and event streaming infrastructure

Web Executive | Student Placement Office (SPO), IIT Kanpur (*Jun'24 – Apr'25*)

- Maintained and enhanced **Next.js** and **Go** applications, supporting **3,000+ active campus users** during placements
- Optimized resume verification workflows by strengthening validation logic and backend processing for placement applications